

UNIT 4: Statistical Data

4.1: No-Code AI for Statistical Data

Lesson Title: No-Code AI for Statistical Data	Approach: Session + Activity
Summary: Understanding the importance and need of No-Code AI for Statistical Data, advantages, and disadvantages of using No-Code AI for Statistical Data, how to use No-Code AI tools for Statistical Data.	
Learning Objectives: • Understand the meaning of No-Code AI and the need of it. • Understand the difference between No-Code and Low-Code. • What are some no-code tools used for the statistical dataset? • How is the AI project cycle different for No-Code AI?	
Learning Outcomes: • Define No-Code and Low-Code AI. • Identify the differences between Code and No-Code AI concerning Statistical Data. • Relate AI project stages to the stages of No-Code AI projects.	
Pre-requisites: Basic knowledge of AI project cycle.	
Key-concepts: • No code tools for AI projects • Understanding different types of approaches like code, low code and No-Code for statistical data. • Identifying and recognizing various tools and platforms for low code and No-Code approaches	

Data science is a journey of exploration and discovery. Artificial Intelligence is a technology which completely depends on data, which is fed into the machine which makes it intelligent. And depending upon the type of data we have; AI can be classified into three broad domains: Data science, Computer Vision and Natural language processing.

Data Sciences is a concept to unify statistics, data analysis, machine learning and their related methods in order to understand and analyze actual phenomena with data. It employs techniques and theories drawn from many fields within the context of Mathematics, Statistics, Computer Science, and Information Science.

What is Data Science ?

Applications of Data Science:

Internet Search: All the search engines (including Google) make use of data science algorithms to deliver the best result for our searched query in the fraction of a second. Considering the fact that Google processes more than 20 petabytes of data every day, had there been no data science, Google wouldn't have been the 'Google' we know today.

Targeted Advertising: If you thought Search would have been the biggest of all data science applications, here is a challenger – the entire digital marketing spectrum. Starting from the display banners on various websites to the digital billboards at the airports – almost all of them are decided by using data science algorithms. This is the reason why digital ads have been able to get a much higher CTR (Call-Through Rate) than traditional advertisements. They can be targeted based on a user's past behaviour.



Website Recommendations: Aren't we all used to the suggestions about similar products on Amazon? They not only help us find relevant products from billions of products available with them but also add a lot to the user experience. A lot of companies have fervently used this engine to promote their products in accordance with the user's interest and relevance of information. Internet giants like Amazon, Twitter, Google Play, Netflix, LinkedIn, IMDB and many more use this system to improve the user experience. The recommendations are made based on previous search results for a user.

Genetics & Genomics:

Data Science applications also enable an advanced level of treatment personalization through research in genetics and genomics. Data science techniques allow integration of different kinds of data with genomic data in disease research, which provides a deeper understanding of genetic issues in reactions to particular drugs and diseases. As soon as we acquire reliable personal genome data, we will achieve a deeper understanding of human DNA. The advanced genetic risk prediction will be a major step towards more individual care.



Activity: Word Scramble the terms related to AI applications.

Purpose: Recall of AI terms

VANAGTOINI APP _____

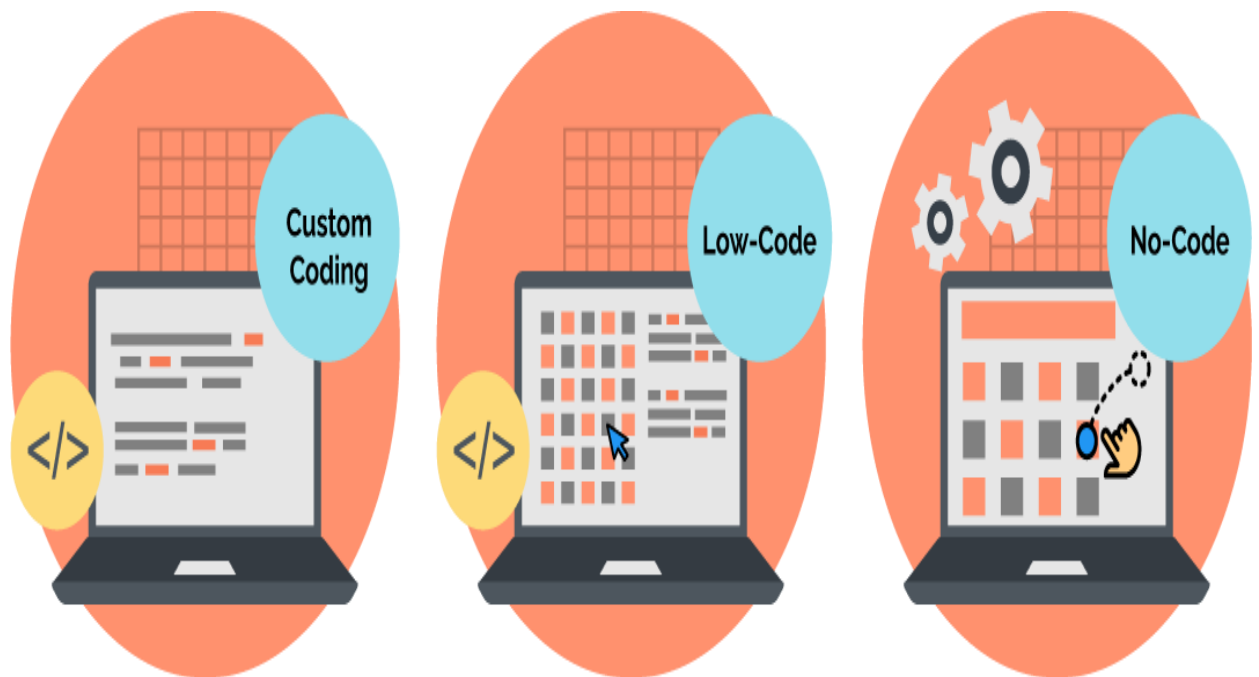
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Introduction to Low/No-Code AI approach for Statistical Data

Let's say you want to build a product, food delivery application. How do you go about starting it?

Building a food delivery application involves several steps, from conceptualization to development, testing, and deployment. The 3 most popular approaches to code are given below.



Custom code is also known as high code.

How do we choose? Which of these 3 is the most suitable for our app?

High code	Low code	No code
High code development refers to traditional software development where programmers write code manually using programming languages like Java, Python, C#, etc.	Low code development involves using platforms or tools that provide visual interfaces and pre-built components to streamline the application development process.	No code development takes low code principles further by allowing users to create applications without any coding or scripting knowledge.
A team of software coders need to write all the code manually.	Programmers need to write some code manually.	Coding knowledge is not required; hence anyone can make the product.
It is expensive.	It is less expensive compared to high code.	It is less expensive compared low code.
The company can own the product they create. You can create anything and customise your product in any way.	You can customise your product to an extent only using code. For example, custom chatbot.	Lack of customisable options as No-Code AI tools are limited to functions in the tool. Simple to use as it uses drag-and-drop features instead of coding.

Now that we have seen the differences,

which approach do you think is the most suitable one for our Food Delivery app? Discuss!

Some inventions that have made life today easier are smartphones, credit cards, internet, onlinestreaming services, Refrigeration technology, GPS navigation, medical innovations etc.

```
models = []
models.append('L1R', LogisticRegression(solver='liblinear', multi_class='ovr'))
models.append('L1A', LinearDiscriminantAnalysis())
models.append('KNN', KNeighborsClassifier())
models.append('ZAR', DecisionTreeClassifier())
models.append('NB', GaussianNB())
models.append('SVM', SVC(gamma='auto'))
# evaluate each model in turn
results = {}
names = []
for name, model in models:
    kfold = StratifiedKFold(n_splits=10, random_state=1, shuffle=True)
    cv_results = cross_val_score(model, X_train, y_train, cv=kfold, scoring='accuracy')
    results.append(cv_results)
    names.append(name)
    print '%s %f' % (name, cv_results.mean(), cv_results.std())
```

```
# Make predictions
from pandas import read_csv
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn import svm

# Load dataset
url = "https://raw.githubusercontent.com/broadml/datasets/master/iris.csv"
names = ["sepal-length", "sepal-width", "petal-length", "petal-width", "class"]
dataset = read_csv(url, names=names)

# Split-out validation dataset
array = dataset.values
X = array[:,0:4]
y = array[:,4]
X_train, X_validation, Y_train, Y_validation = train_test_split(X, y, test_size=0.20, random_state=0)

# Make predictions on validation dataset
model = svm.SVC(gamma='auto')
model.fit(X_train, Y_train)
predictions = model.predict(X_validation)

# Evaluate predictions
print(accuracy_score(Y_validation, predictions))
print(confusion_matrix(Y_validation, predictions))
print(classification_report(Y_validation, predictions))
```

more code to pick the best algorithm...

```
# evaluate each model in turn
results = []
names = []
for name, model in models:
    kfold = StratifiedKFold(n_splits=10, random_state=1, shuffle=True)
    cv_results = cross_val_score(model, X_train, Y_train, cv=kfold, scoring='accuracy')
    results.append(cv_results)
    names.append(name)
    print('%s: %f (%f)' % (name, cv_results.mean(), cv_results.std()))

# Compare Algorithms
pyplot.boxplot(results, labels=names)
pyplot.title('Algorithm Comparison')
pyplot.show()
```

That's a lot of code, right?
And that's why we have No-Code AI.

- In No-Code AI, we can drag and drop, these models in few seconds.
- No coding knowledge is required to implement complex ML algorithms
- Drag and drop feature of a No-Code tool makes it easier.



Why do we need No-Code AI?

- We tend to run into many types of errors when we are coding, and it can be very troublesome at times.
- In No-Code AI since we do not need to code, we won't have any code errors!
- No-Code AI helps to save cost for businesses as it is costly to implement completely coded AI systems.
- Companies can implement AI with less stress and without the need to hire an AI staff with No-Code AI.
- No-Code AI is easy to use – even middle school students can create AI using No-Code tools
- Since it has visual & drag-and-drop features, anyone can see what they are building in real-time

Who can use No-Code AI?

- No-Code AI makes AI more accessible to the general public.
- Non-technical people such as doctors, architects, musicians may quickly construct accurate AI models with no coding involved.

Let's look at a scenario to understand who can use No-Code AI

- No-Code AI makes AI more accessible to the general public.
- Non-technical people such as doctors, architects, musicians may quickly construct accurate AI models with no coding involved.

Thus No-Code AI can empower individuals and organizations across various industries and skill levels to harness the potential of artificial intelligence for their specific needs.

Let's look at a scenario to understand who can use No-Code AI.



Problem: Kayla is a wildlife animal's dietitian manager at the zoo. She takes care of the cost of buying meat and vegetables for animals. With the prices of food increasing rapidly, it will become more expensive for the zoo to buy healthy and nutritious foods for its animals. Therefore, the zoo's accounts team wants to know the increase in the price of food so that they can ask the government or sponsors to fund for the food. Thus, Kayla requires the help of AI to predict the price.

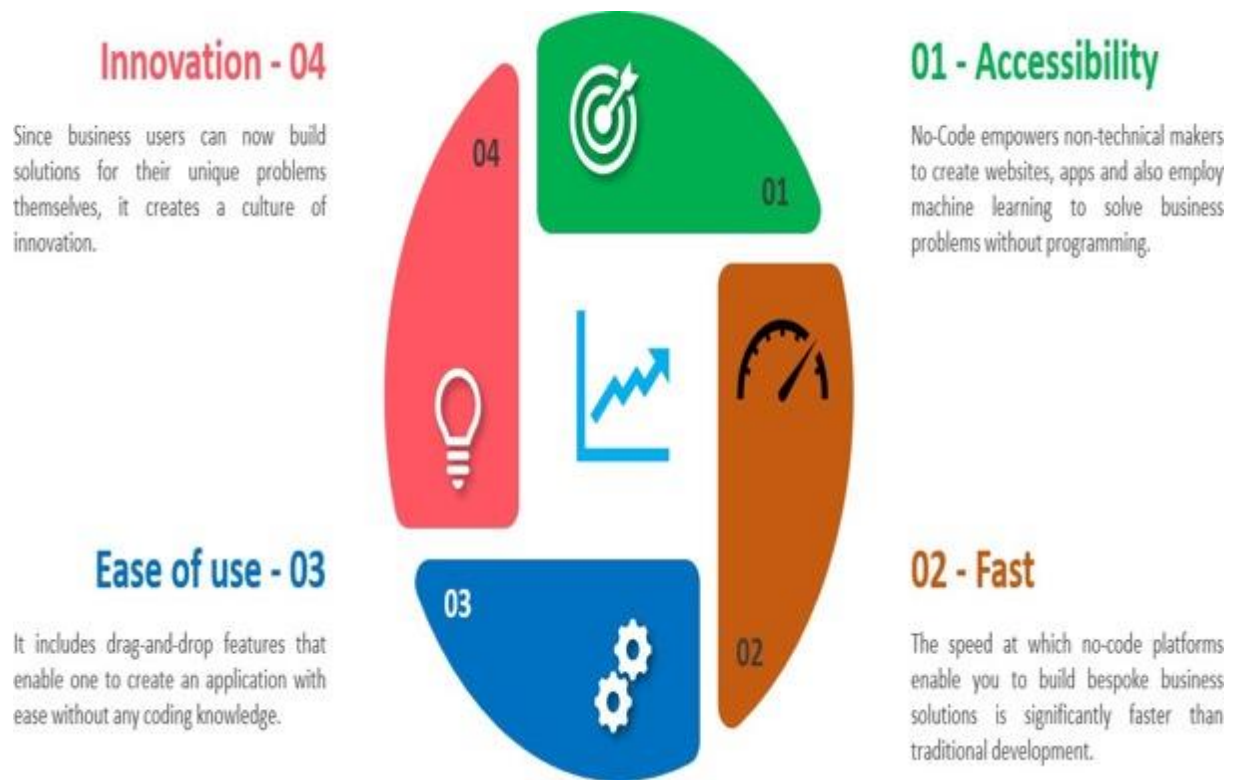
However, Kayla does not know how to code to build an AI model as she has not learnt AI before.

Solution:

She finds a No-Code AI tool online that can help her build an AI model to predict the prices of meat and vegetables without any code!

Kayla can also use any popular No-Code Tool like Orange Data Mining to build a price prediction model. We will look at how to build a simple price prediction model in a short while. From this scenario you can infer that: You do not need to know coding to build an AI model and anyone can make use of No-Code AI to build an AI model.

Benefits of No-Code Tools



Now that we have discussed the advantages of No-Code tools, let us discuss the disadvantages too...

Disadvantages of No-Code Tools

Lack of Flexibility

Drag-and-drop elements can be very convenient. On the other hand, you are limited to those fixed elements. Hence, you have a limitation in customizing your application in no-code platforms, and it does not give you all the authority and flexibility you might need.

Automation Bias

Automation bias is the tendency for humans to favor suggestions from automated decision-making systems and to ignore contradictory information made without automation, even if it is correct.

Security Issues

No code platforms do not essentially force you to think of security first or even evaluate security best practices. Therefore, these applications are only best suited for companies that don't deal with sensitive data, as these platforms usually offer limited control over it.

Popular No-Code Tools

Here are some of the top No-Code AI tools where we can build our AI models in.

No-Code tool	Details	Released
Azure Machine Learning	Cloud-based service provided by Microsoft	July 2014
Google Cloud AutoML	Cloud-based service provided by Google	January 2018
Orange Data Mining	An open-source data visualization, machine learning and data mining toolkit. Developer: University of Ljubljana	October 1996
Lobe AI	Lobe AI is a machine learning platform that enables to create custom machine learning models using a visual interface	2015
Teachable Machine	Teachable Machine is a web-based tool that makes creating machine learning models fast, easy, and accessible to everyone.	November 2017

Azure Machine Learning

Microsoft's Azure Machine Learning aims to simplify ML processes.

- It allows users to build ML models without having to touch any code,
- and provides an easy interface for cleaning up your data, training models, evaluating them and, finally, putting them into production.



Google Cloud AutoML

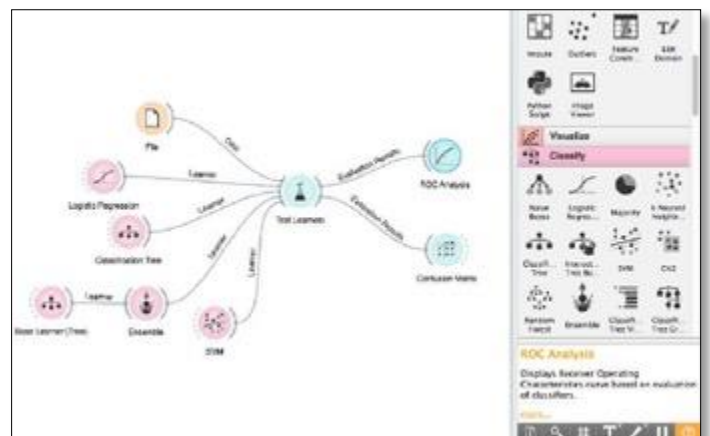
- Users with limited ML knowledge can train high-quality models specific to their business needs with minimal effort.
- They can build their own custom machine learning model in minutes and then use the models in their applications and web sites.

Click on [Introducing Cloud AutoML](#) to watch a video on it.



Orange Data Mining

- Orange enables you to visualize data and perform data mining and machine learning.
- You can use this tool without writing a single code.
- This platform can be used for analysis, is relatively easy, and has beautiful visuals.



Now, let us get back to building a simple prediction model using Orange Data Mining for Kayla!

FAO food price index							
		Food Price Index ¹	Meat ²	Dairy ³	Cereals ⁴	Vegetables Oils ⁵	Sugar ⁶
2004		65.6	67.6	69.8	64.0	69.6	44.3
2005		67.4	71.8	77.2	60.8	64.4	61.2
2006		72.6	70.5	73.1	71.2	70.5	91.4
2007		94.3	76.9	122.4	100.9	107.3	62.4
2008		117.5	90.2	132.3	137.6	141.1	79.2
2009		91.7	81.2	91.4	97.2	94.4	112.2
2010		106.7	91.0	111.9	107.5	122.0	131.7
2011		131.9	105.3	129.9	142.2	156.5	160.9
2012		122.8	105.0	111.7	137.4	138.3	133.3
2013		120.1	106.2	140.9	129.1	119.5	109.5
2014		115.0	112.2	130.2	115.8	110.6	105.2
2015		93.0	96.7	87.1	95.9	89.9	83.2
2016		91.9	91.0	82.6	88.3	99.4	111.6
2017		98.0	97.7	108.0	91.0	101.9	99.1
2018		95.9	94.9	107.3	100.8	87.8	77.4
2019		95.1	100.0	102.8	96.6	83.2	78.6
2020		98.1	95.5	101.8	103.1	99.4	79.5
2021		125.7	107.7	119.1	131.2	164.9	109.3
2021	September	129.2	112.7	118.1	132.8	168.6	121.2
	October	133.2	112.0	121.5	137.1	184.8	119.1
	November	135.3	112.5	126.0	141.4	184.6	120.2
	December	133.7	111.0	129.0	140.5	178.5	116.4
2022	January	135.6	112.1	132.6	140.6	185.9	112.7
	February	141.2	113.9	141.5	145.3	201.7	110.5
	March	159.7	119.3	145.8	170.1	251.8	117.9
	April	158.4	121.9	146.7	169.7	237.5	121.5
	May	158.1	122.9	144.2	173.5	229.2	120.4
	June	154.7	125.9	150.2	166.3	211.8	117.3
	July	140.6	124.1	146.5	147.3	168.8	112.8
	August	137.9	122.0	143.4	145.6	163.3	110.5
	September	136.3	121.4	142.5	147.8	152.6	109.7

Purpose: To build an AI model to predict price using Orange Data mining AI tool

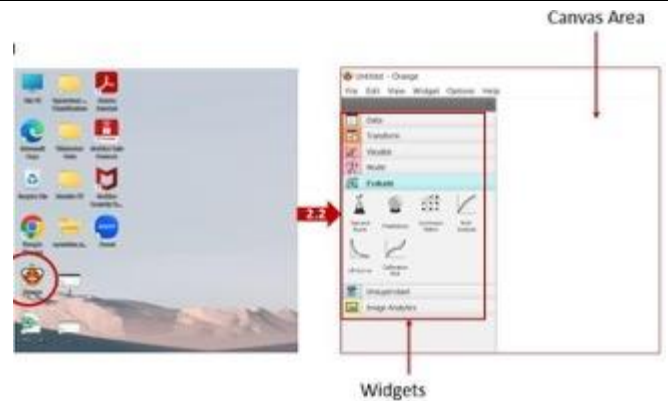
Steps for building a simple priceprediction model:

Step 1: Download the datasetfrom

<https://www.fao.org/worldfoodsituation/foodpricesindex/en/>

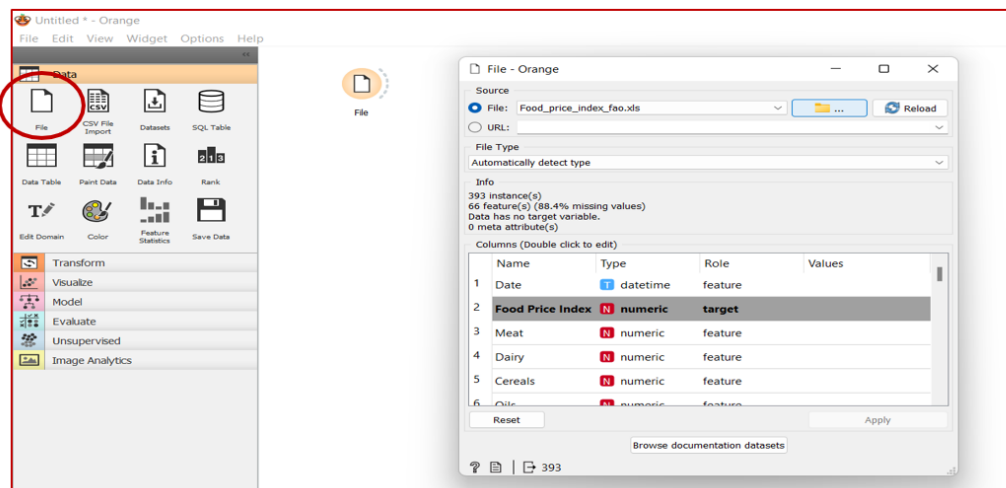
Step 2: Open Orange Data Mining

- Double Click on Orange Icon
- Open the tool



Step 3: Upload the Dataset

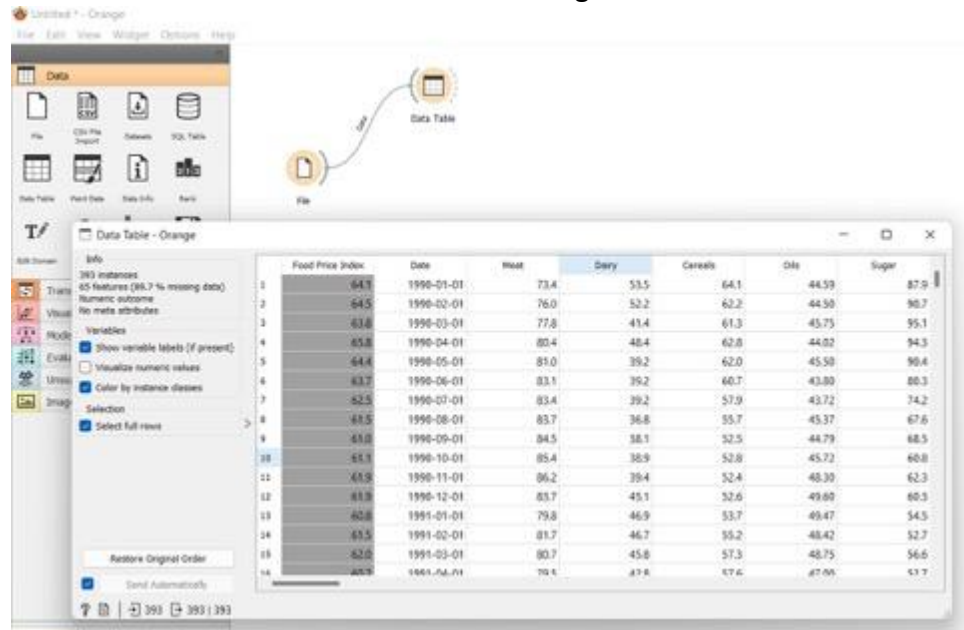
- Click on the File widget under Data Menu
- File widget will appear on the canvas
- Click on it and browse to the folder to upload the dataset
- Select the target variable as Food Price Index, as we are trying to predict the price index.



Step 4: View the Dataset

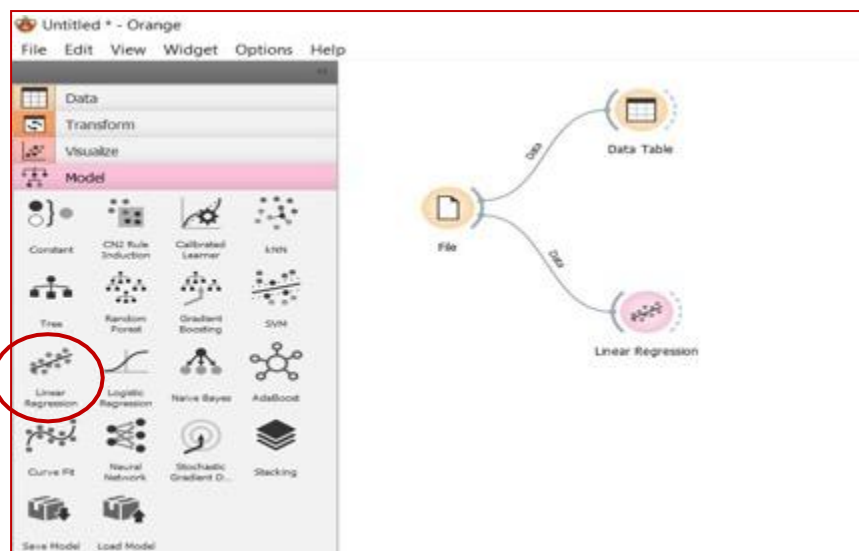
- Click on the Data Table widget under Data Menu
- Data Table widget will appear on the canvas
- Connect File to Data Table

- Click on Data Table to view the dataset using the tool.



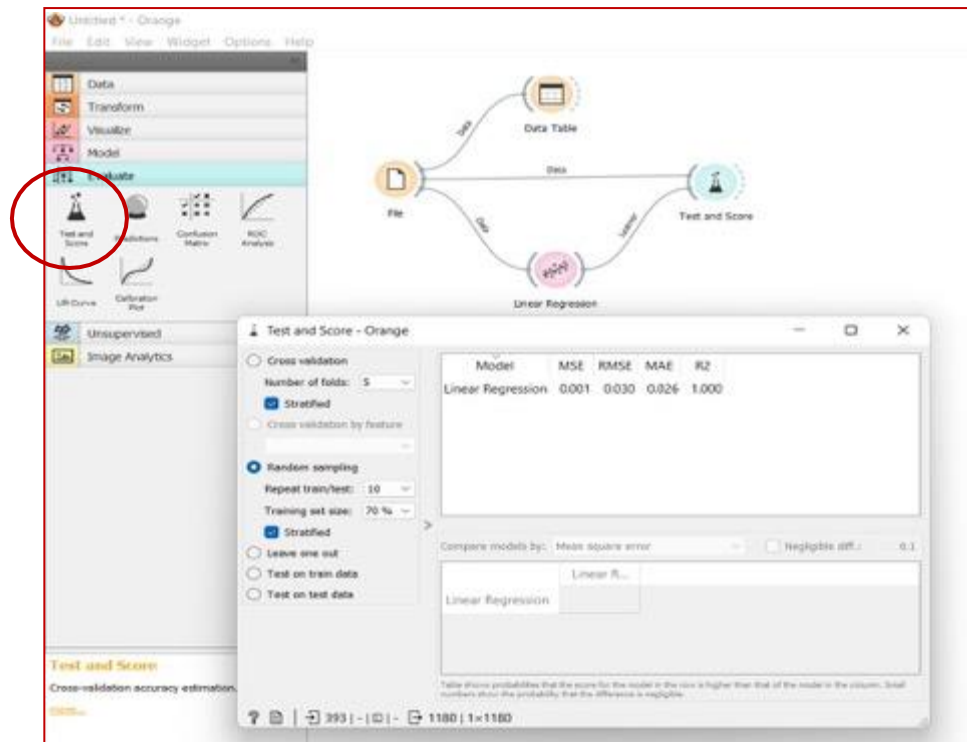
Step 5: Select the Model for Prediction

- Click on the Linear Regression widget under Model Menu
- Linear Regression widget will appear on the canvas
- Connect File to Linear Regression
- Linear Regression is an algorithm used for Regression Model



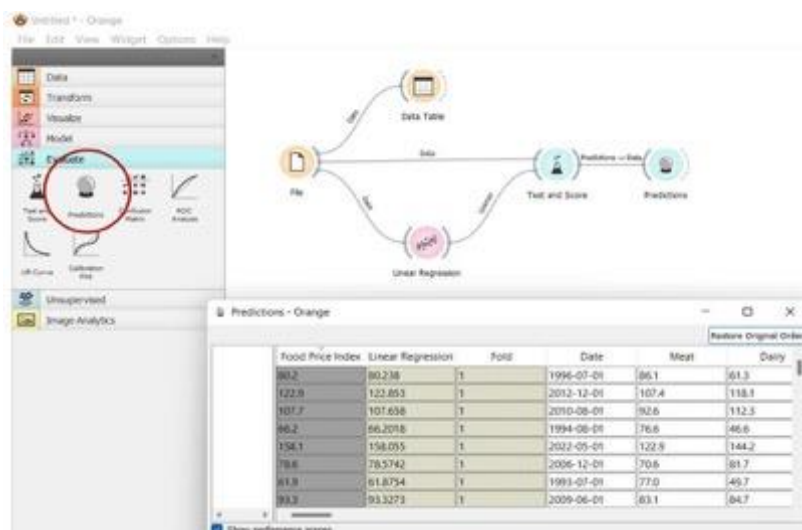
Step 6: Evaluate the Model

- Click on the Test and Score widget under Evaluate Menu
- Test and Score widget will appear on the canvas
- Connect File and Linear Regression to Test and Score to check the performance parameters.
- Click on Test and Score to view the parameters.



Step 7: Model Prediction

- Click on the Prediction widget under Evaluate Menu
- Prediction widget will appear on the canvas
- Connect the Test and Score to the Prediction to check the prediction made by Logistic Regression Model.
- Click on Prediction to view the price prediction.



Kayla could use such a model to make the price prediction for animal feed and make a systematic fund-raising plan for the Zoo.

Other No code AI tools

Lobe



- Machine Learning has been made easy using Lobe.
- It has everything that you need to bring your machine-learning ideas to life.
- Lobe helps you, train models with a free and easy-to-use tool.
- It automatically trains a custom machine-learning model that can be shipped in your app.

Teachable Machine



- Teachable Machine is a web-based tool that makes creating machine learning models fast, easy and accessible to everyone.
- It can be used to train a computer to recognize your own images, sounds and poses.
- No specific expertise or coding is required to use this tool.

Test Yourself:

1. Orange data mining is an example of

- a) Custom coding
- b) Low code
- c) No code
- d) None of these.

2. Select Which is not the feature of No code approach.

- a) visual
- b) Highly expensive
- c) code free
- d) drag and drop

3. _____ assembly relies on developers to write and deploy code.

- a) High code
- b) Low code
- c) No code
- d) None of these

c) 4. Flexibility is often limited in _____

- a) High code c) No code
- b) High code and Low code d) High code and No code.
5. The organisation is heavily dependent on developer resources. This statement is true for
- a) High code approach c) No code approach
- b) Low code approach d) All of the above.

Answer the following briefly:

1. Name any two-cloud based No-code AI tools?
2. How accessible are no-code AI tools for non-technical users?
3. What types of tasks can be accomplished with no-code AI tools?
4. Can no-code AI tools be used for advanced projects? Justify.
5. Do no-code AI tools require prior programming knowledge? Justify.

Answer the following in detail:

1. What are the benefits of using No-code AI tools?
2. What are the challenges faced in using the No-code AI tools?
3. Differentiate low code and No-code AI tools with examples.
4. As the CEO of a small e-commerce startup, you're eager to leverage artificial intelligence to enhance your platform's user experience and drive sales. However, your team lacks the technical expertise to develop and deploy AI-powered solutions. What would be your recommendations for the CEO?
5. Samarth attended a seminar on Artificial Intelligence and has now been asked to write a report on his learnings from the seminar. Being a non-technical person, he understood that the AI enabled machine uses data of different formats in many of the daily based applications but failed to sync it with the right terminologies and express the details. Help Samarth define Artificial Intelligence, list the three domains of AI and the data that is used in these domains.

Reflection Time:

- 1.Would you like to explore the tools specified here to build No-Code AI projects? How?
- 2.Are you aware of any other tools or platforms used for low code or No-Code AI Projects? List them.